Project Proposal

Project name: Self-Supervised Depth Estimation from Monocular Edge Vision

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Course:ECA - Edge Computing Applications

1. Introduction and Problem Statement

The aim of this project is to build a system that estimates depth from a single image. The output will be a depth map, displayed using colours to show which parts of the scene are closer and which are farther away.

The main challenge is to produce useful depth predictions with a small model that can run on the NXP i.MX 8M Plus board. Larger models may require too much computation and memory, while smaller models may lose accuracy and detail.

We will use a **teacher–student** framework. A pretrained stereo teacher will generate predictions using a left and a right image of the same scene. A smaller monocular student will learn from these predictions while receiving only one image. This follows the research direction specified in the project description.

2. Proposed Solution and the Role of Edge AI

The teacher will be an existing pretrained stereo Transformer.The teacher will be selected from the GMStereo models, according to the project requirements and available computing resources. We will not train the teacher from scratch.
GMStereo models:
https://github.com/autonomousvision/unimatch/blob/master/MODEL_ZOO.md

The student will use MobileNetV3 as its feature-extraction backbone, with an additional decoder that produces the depth prediction. Training will take place on a computer. After training, only the student will run on the NXP board. The teacher and the second image will not be needed during final use.

The Edge AI component is the local execution of the student on the board. Our implementation will aim to use the board's NPU, and we will report which parts of the model are accelerated successfully.

3. Challenges in Existing Practices

The selected literature highlights several challenges:

- Collecting accurate depth labels can be difficult. Self-supervised methods instead use relationships between images as a training signal.
- Image-reconstruction methods can produce unreliable results in hidden regions and areas where image matching is difficult.
- Models intended for embedded use must balance prediction quality with processing time and memory requirements.
- Results obtained on one hardware platform do not establish performance on another platform.

Monodepth and Monodepth2 address learning from image relationships, while FastDepth and Lite-Mono investigate efficient depth estimation. These approaches provide the background for our project.

4. Project Scope and Goals

1. Prepare separate training, validation and test data.
2. Generate teacher predictions and exclude unreliable training targets.
3. Build a MobileNetV3-based student with a depth decoder.
4. Train the student with the teacher guidance.
5. Convert the trained student for deployment on the NXP board.
6. Measure prediction accuracy, model size, memory use and processing speed.

6. Team Roles and Responsibilities

Team member	Main responsibilities
Andreea	Prepare the data, run the teacher, save predictions, implement training losses, train the student with guidance and evaluate depth accuracy.
Simona	Train the student without guidance and evaluate depth accuracy, check the board environment, test model export, create the demonstration and measure runtime performance.
Both	Review code, integrate components, test the complete system, review the selected papers, and prepare the report and presentation.

8. Literature Review

Monodepth learns depth from stereo images without ground-truth depth labels, using image reconstruction as a training signal. Monodepth2 improves this approach by reducing the

influence of hidden regions and unreliable pixels. These methods support the **self-supervised** part of our project.

PackNet-SfM focuses on preserving fine spatial details, while FastDepth and Lite-Mono explore efficient depth-estimation models. They highlight the need to balance prediction quality, processing speed and memory use on edge devices.

Depth Anything uses teacher-generated predictions to support learning from unlabelled images. This motivates teacher guidance, although its approach is not the stereo-teacher setup required by our project. The notes on resource-constrained devices also emphasise quantization and hardware compatibility.

Our project will investigate whether a stereo teacher improves a compact monocular MobileNetV3 student. We will compare training with and without teacher guidance, then evaluate accuracy and performance after quantization and deployment on the NXP i.MX 8M Plus.

Selected Papers

- Godard et al. (2017). *Unsupervised Monocular Depth Estimation with Left-Right Consistency*.
- Godard et al. (2019). *Digging Into Self-Supervised Monocular Depth Estimation*.
- Wofk et al. (2019). *FastDepth: Fast Monocular Depth Estimation on Embedded Systems*.
- Zhang et al. (2023). *Lite-Mono: A Lightweight CNN and Transformer Architecture for Self-Supervised Monocular Depth Estimation*.
- Yang et al. (2024). *Depth Anything: Unleashing the Power of Large-Scale Unlabeled Data*